A) OM-85

Epithelial barrier modulation

- ↑IFN-β and IFN-γ
- ↓ viral-induced cell death
- ↓ ACE2, TMPRSS2 and ICAM-1
- ↑ β-defensin
- ↑ nitric oxide
- ↑ ciliary beat frequency
- ↑ transepithelial resistance (TEER)
- ↓ mucus production

Innate immunity activation

- Activation Macrophages and DCs:
  - ↑ cytokines: TNF-α, IL-6, IFN-β, IL-12p70
  - ↑ chemokines: IL-8, CCL2, CCL20
- Bone-marrow-derived macrophages: ↑ NO
- DCs maturation: ↑ MHC-II, CD80, CD86, CD83

Adaptive immunity modulation

- T cell response:
  - ↑ Th1 responses: IFN-γ-producing $CD4^+$ T cells and $CD8^+$ T cells
  - ↓ Th2 responses: IL-4, IL-5, IL-13
- B cells activation: ↑ IgA and IgG

Inflammation control

- Prevention of excessive activation of inflammasome
- ↑ Treg and tolerogenic DCs
- ↓ inflammatory eosinophils
- Modulation of IL-33/ILC2 axis*

Preclinical evidence +++

B) PMBL

Epithelial barrier modulation

- ↑ E-cadherin
- ↑ amphiregulin
- ↑ β-defensin-2
- ↑ ICAM-1
- ↑ IL-23

Innate immunity activation

- Activation macrophages and DCs:
  - ↑ transcription of cytokines, chemokines and alarmins: *CXCL1, CCL20, IL-6, TNF-α, S100A9, LCN2*
- DCs maturation: ↑ MHC-II, CD80, CD86, CD83
  - ↑ cytokines: IL-12p70 and IFN-γ

Adaptive immunity modulation

- T cell responses:
  - Th1 response: ↑ IL-2, IL-10, IFN-y
  - ↑ CD25 expression on B cells, $CD4^+$ T cells and $CD8^+$ T cells and ↑ IL-2/IL-2R
- B cell activation: ↑ IgA and IgG

Inflammation control

- Th2 profile ($CD4^+$ T cells): ↑ IL-4, IL-10,
- Th1 profile ($CD8^+$ T cells): ↑ IFN-y and IL-2

Preclinical evidence +

C) MV130

Epithelial barrier modulation

- ↑ occludin and zonula occludens-1

Innate immunity activation

- Activation macrophages and DCs:
  - ↑ cytokines: IL-12p70, TNF-α, IL-6, IL-1β and IL-23
- Trained immunity: epigenetic and metabolic reprogramming

Adaptive immunity modulation

- T cell response:
  - $CD4^+$ T cells: ↑ Th1 response, ↑ Th17 response
  - ↓ Th2 cytokines: IL-5, IL-13
- B-cell activation : ↑ IgA and IgG*

Inflammation control

- ↑ IL-10 production
- ↓ inflammatory eosinophils*
- Macrophage polarization: M2 phenotype

Preclinical evidence +

D) CRL1505

Epithelial barrier modulation

- Gut-lung axis:
  - ↓ barrier permeability, preserving epithelial integrity

Innate immunity activation

- Activation macrophages and DCs :
  - ↑ cytokines: IFN-α/β, IFN-γ, IL-12p70
- DCs maturation: ↑ MHC-II, CD86,

Adaptive immunity modulation

- T cell response:
  - ↑ Th1 responses
- B-cell activation : ↑ IgA and IgG

Inflammation control

- ↑ IL-10, IFN-γ

Preclinical evidence ++